

JEE MAIN REVISION MASTER: CHEMICAL KINETICS AND NUCLEAR CHEMISTRY SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 221 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Rate Law, Order & Integrated Rate Equations	147 PYQs	Rate Laws
Very High	Arrhenius Equation & Activation Energy	82 PYQs	Temperature & Catalysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Rate Law, Order & Integrated Rate Equations	$\text{Zero: } [A] = [A]_0 - kt, \quad t_{1/2} = \frac{[A]_0}{2k}, \quad \text{First: } k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}, \quad t_{1/2} = \frac{0.693}{k}$
Arrhenius Equation & Activation Energy	$k = A e^{-E_a/RT}, \quad \ln k = \ln A - \frac{E_a}{RT}, \quad \log_{10} \left(\frac{k_2}{k_1} \right) = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_2 T_1} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Rate Law, Order & Integrated Rate Equations (Very High Yield — 147 PYQs)

Core Principle: Differential and integrated rate laws, half-life formulas, and overall reaction order determined from initial rates.

■ **Caution / JEE Pitfall:** For a first-order reaction, half-life $t_{1/2} = (0.693 / k)$ is strictly INDEPENDENT of initial concentration $[A]_0$.

■ **Speed Trick:** For an n^{th} order reaction: $t_{1/2} \propto \frac{1}{[A]_0^{n-1}}$. Units of rate constant $k = \text{mol}^{1-n} \text{L}^n \text{s}^{-1}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$ and $1 \text{ cal} = 4.2 \text{ J}$ Statement I : When $E_a = 12.6 \text{ kcal/mol}$, the room temperature rat..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Consider the reaction $aX \rightarrow bY$, for which the rate constant at 30°C is $1 \times 10^{-3} \text{ mol}^{-1} \text{ L s}^{-1}$. Which of the ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): " $A \rightarrow \text{product}$ (First order reaction). Three sets of experiment were performed for a reaction under similar experimental conditions..." [Ans: N/A]

3.2 Arrhenius Equation & Activation Energy (Very High Yield — 82 PYQs)

Core Principle: Temperature dependence of reaction rates, activation energy E_a , Arrhenius plot ($\ln k$ vs $1/T$), and catalyst energy profiles.

■ **Caution / JEE Pitfall:** A positive catalyst increases rate by LOWERING activation energy E_a ; it does NOT alter the equilibrium constant K_{eq} or ΔH !

■ **Speed Trick:** Plot of $\log_{10} k$ vs $(1/T)$ gives a straight line with slope = $-(E_a / 2.303 R)$ and y-intercept = $\log_{10} A$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "At 27°C in presence of a catalyst, activation energy of a reaction is lowered by 10kJmol^{-1} . The logarithm of ratio of $\frac{k_{\text{cat}}}{k_{\text{uncat}}}$ is..." **[Ans: A]**
- **[2026 Evening Shift] Q#1 (Numerical):** " $A \rightarrow B$ (first reaction) $C \rightarrow D$ (second reaction) Consider the above two first-order reactions. The rate constant for first reaction at 500 K is..." **[Ans: 5]**
- **[2026 Morning Shift] Q#1 (Numerical):** "The temperature at which the rate constants of the given below two gaseous reactions become equal is ____ K. (Nearest integer)" **[Ans: 1303]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 24th January Morning Shift (Q#1)

At 27°C in presence of a catalyst, activation energy of a reaction is lowered by 10kJmol^{-1} . The logarithm of ratio of $(k_{\text{catalysed}} / k_{\text{uncatalysed}})$ is.... (Consider that the frequency factor for both the reactions is same)

- (A) 1.741 **[Correct]**
- (B) 0.1741
- (C) 17.41
- (D) 3.482

Correct Answer: A

Solution Analysis: Arrhenius equation (NCERT): $k = Ae^{-\frac{E_a}{RT}}$ Given: catalyst lowers activation energy by $10\text{kJmol}^{-1} = 10000\text{Jmol}^{-1}$ and frequency factor A is same. So,
 $\frac{k_{\text{cat}}}{k_{\text{uncat}}} = \frac{Ae^{-\frac{E_{a,\text{cat}}}{RT}}}{Ae^{-\frac{E_{a,\text{uncat}}}{RT}}} = e^{\frac{E_{a,\text{uncat}} - E_{a,\text{cat}}}{RT}} = e^{(10000 / RT)}$
 At 27°C , $T = 300\text{K}$ and $R = 8.314\text{Jmol}^{-1}\text{K}^{-1}$: $(10000 / RT) = (10000 / 8.314 \times 300) \approx 4.009$ So,
 $\log_{10}\left(\frac{k_{\text{cat}}}{k_{\text{uncat}}}\right) = \log_{10}(e^{4.009})$

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

If the half life of a first order reaction is 6.93 minutes then the time required for completion of 99% of the reaction will be ____ minutes. (Given : $\log 2 = 0.3010$)

Correct Answer: 46

Solution Analysis: For a first order reaction, $t_{1/2} = (0.693 / k)$ Given: $t_{1/2} = 6.93\text{min}$ So, $k = (0.693 / 6.93) = 0.1\text{min}^{-1}$
 Now, 99% completion means only 1% of the reactant is left. So, $\frac{[A]_0}{[A]} = (100 / 1) = 100$ For a first order reaction,
 $t = (2.303 / k) \log \frac{[A]_0}{[A]}$ Substituting the values: $t = (2.303 / 0.1) \log 100$ Since, $\log 100 = 2$ therefore, $t = (2.303 / 0.1) \times 2$
 $t = 23.03 \times 2 = 46.06\text{min}$ Hence, the time required for 99% completion of the reaction is 46.06 minutes